



DEALER BEST PRACTICE SERIES

On-board Fluid Filtration

Application Maintenance Site Management Component Rebuild **Component Life Management** MARC Management

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1.0 Introduction

Maintaining very clean fluid throughout the entire life cycle of the major component or system is the best way to maximize component life and improve reliability. The micron rating of filters on many mining machine systems are sized so as not to plug during the initial break-in period on new machines. This does not provide optimal filtration capability to maintain very high levels of oil cleanliness after the break-in period is complete.

2.0 Best Practice Description

Ultra-High Efficiency (UHE) 6-micron hydraulic filters are the most aggressive approach to removing break-in debris as quickly as possible and maintaining the highest level of fluid cleanliness. The UHE filters are used in place of standard filters for all machine systems except the engine. Because these filters effectively trap very small particles, some filter plugging will occur during the initial PM periods. The frequency and duration of filter plugging will vary by system.

2.1 Break-in Debris

2.1.1 Steering System

The steering systems essentially consist of a small piston pump, steering cylinders and a control valve. Once the initial assembly debris is removed from this system, it produces very little break-in debris and normally stays very clean when a UHE filter is installed.

2.1.2 Transmission

Transmission clutch discs, and gears produce moderate amounts of break-in debris. This process usually takes between 100-200 hours to complete. Even if the transmission were perfectly clean after assembly, break-in debris would still be produced. When standard filters are replaced by UHE filters on transmission and torque converter circuits, plugging will occur during the break-in period of 150-200 hours. The vast majority of machines will plug only one filter. A few will plug a second filter. After the break-in period is complete, the UHE filters will usually maintain the fluid at ISO 16/13 at all times with no further plugging.

2.1.3 Final Drive & Differential

The rear axle produces very large amounts of break-in debris from gears. This process may last up to 8,000 hours. This debris is almost all very small abrasive particles from the hardened gears. Due to inconsistent performance of the rear axle lube filter on machines with ground driven lube pumps, it is difficult to estimate the frequency and duration of filter plugging. The new 793D has a hydraulically driven continuous lube system that is much more effective. However, there is virtually no experience with this system yet.

Improved on-board machine filtration generally consists of three steps:

2.2 Ultra High Efficiency (UHE) Filters

Install 6-micron UHE filters wherever possible.

2.3 Supplemental Filters

Any hydraulic circuit or component that has the opportunity to contaminate the hydraulic tank with debris in the event of a failure should have supplemental filters installed. This includes

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hydraulically operated fan motors that often have no filtration, and the case drain lines on some hydraulic pumps.

2.4 Off-Board Filtration

Off-board filtration should be used to supplement the UHE filters in the rapid removal of break-in debris on new machines or with rebuilt components or major system repairs. Particle counts should be taken at the end of the PM period before PM service. When the UHE filters can maintain ISO 18/15 or better throughout the PM interval without plugging for two consecutive PM intervals, they are capable of maintaining cleanliness. Off-board filtration may be discontinued at this point.

3.0 Implementation Steps

Three basic steps are required:

3.1 UHE Filter Part Numbers

Determine the machines models and identify the part number and quantity of UHE filters required. Contact Matt Kimbro, KIMBRO_MATT_G@cat.com, Caterpillar Filters & Fluids Group.

3.2 Off-Board Filter Carts

The use of off-board filtration is required after major repairs and recommended to accelerate the completion of the component break-in period. An adequate number of properly sized carts and tooling is required for repair and PM.

3.3 Customer Understanding and Acceptance

Both customer and dealer personnel must understand and accept that premature filter plugging is expected for a short period of time. When filter plugging occurs, the machine should be brought in, a new filter quickly installed, and the machine put back into service immediately. Whether the machine is brought in immediately or at the end of the shift is at the discretion of the dealer and customer.

If the customer or dealer is not willing to accept filter plugging during the break-in period, this initiative should not be implemented.

4.0 Benefits

Maintaining oil cleanliness for major components and machine systems increases both reliability and durability. Electro-hydraulic control valves, which are widely used in transmission controls and implement hydraulics, are very intolerant of microscopic, ferrous debris. Heavily loaded wheel and final drive bearings, as well as, duo-cone seals are also easily damaged by abrasive debris from break-in of gears. Quickly reaching and maintaining ISO 18/15 or better significantly reduces the number of contamination induced failures and repairs and significantly extends component life. This is the most effective way to reduce cost per ton and increase reliability and durability.

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5.0 Resources Required

The number and size of filter carts required is determined by fleet size. Carts may be purchased through Cat Service Tool Group, outside suppliers, or built by the dealer. Training for maintenance and operation personnel is also required so that they fully understand the function and importance of the use of filter carts.

6.0 Supporting Attachments

[Component Life Management Master Document](#) PDF (Click on “Attachments” tab inside of this document to open/view)



[“Improving Component Durability”](#) booklet set - Form # SEBF1021, (see next page)

Consists of one of each of the following:

Fuel Systems	SENR9620
Final Drives and Differentials	SEBF1015
Powershift Transmissions	SEBF1016
Component R&I	SEBF1017
Engines	SEBF1018
Hydraulics	SEBF1019
Managing Fluid Cleanliness	SEBF1020

7.0 Related Best Practices

0806-2.10-1003	-Managing Fluid Cleanliness
0806-2.10-1002	-Off-board Fluid Filtration

8.0 Acknowledgements

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